GPSC CBRT · HOMOEOPATHIC PHYSICIAN · CORE

Topic 6: Pathology, Microbiology & Diagnostics

Complete notes with pictures, memory hooks, rapid revision and practice MCQs.

Topic 6: Pathology, Microbiology & Diagnostics

How to use this in 2 days

Topic 6 is one of ten core topics, so expect roughly 2–4 of the 25 core marks from it. GPSC asks one-line recall: a stain, an organism, a vector, a normal value, a test of choice. Learn the tables, not the prose.

Session	Read	Then do
Day 1 morning	GPSC questions already asked; cell injury; inflammation; haemodynamics	Mark every fact you got wrong
Day 1 afternoon	Neoplasia; immunity and amyloid; general microbiology	First 25 practice MCQs
Day 1 evening	Bacteria; viruses	Rapid revision, sections 1–6
Day 2 morning	Parasites and fungi; haematology	—
Day 2 afternoon	Clinical pathology (LFT, RFT, urine, CSF); imaging and serology	Last 25 practice MCQs
Day 2 evening	Rapid revision page twice; your marked errors	Sleep

MCQ habits that pay here:

- **'Most common' and 'investigation of choice'** are the favourite stems. Each table below has a column for them.
- **Stains and media** are pure recall: learn the pairs (Ziehl–Neelsen → AFB, Giemsa → malaria, India ink → Cryptococcus).
- **Normal values**: learn one number per test, not ranges from every book.
- **Goa context**: leptospirosis, scrub typhus, dengue, malaria and KFD-type monsoon fevers come up because they are local.

What GPSC has already asked

- GPSC's Medical Officer papers of 28 Sep 2024 and 28 Sep 2025 asked about 17 of their 60 core questions from this topic's ground.
- Almost all of them test a single linked fact: a sign and its cancer, a test of choice, a radiology sign, an organism.
- The answers below are GPSC's keys, checked.

Paper	Question (shortened)	Answer	The fact to carry away
MO 2025	Rat-tail appearance on barium swallow	Carcinoma oesophagus	Rat tail / shouldering = carcinoma; bird's beak = achalasia
MO 2025	Best diagnostic modality for TB spine	CT-guided biopsy	Tissue (histology + culture/CBNAAT) confirms; MRI is best imaging
MO 2025	Whitish tonsillar membrane that bleeds on scraping	Diphtheria	Corynebacterium diphtheriae; Albert stain, Loeffler's serum slope, tellurite medium
MO 2025	Suspected COVID-19 with loss of smell: next step	Nasopharyngeal swab for RT-PCR	NP swab RT-PCR is the reference test
MO 2025	Investigation of choice, acute subarachnoid haemorrhage	Non-contrast CT	CT first; lumbar puncture (xanthochromia) if CT negative
MO 2025	Screening method for CIN — the exception	Colposcopic biopsy	Pap smear, HPV-DNA and VIA screen; biopsy diagnoses
MO 2025	Intraocular metallic foreign body: first investigation	Thin-cut CT orbit	MRI is contraindicated for metallic foreign bodies
MO 2024	Paget disease of the nipple is	Malignant	Underlying ductal carcinoma; large clear Paget cells in epidermis
MO 2024	Lesion with higher risk of invasive breast cancer	Atypical ductal hyperplasia	Atypical hyperplasia ≈ 4–5× risk; sclerosing adenosis, papillomatosis lower
MO 2024	Troisier's sign	Carcinoma stomach	Left supraclavicular node (Virchow's node)
MO 2024	Leukonychia is seen in	Hypoalbuminaemia	Koilonychia = iron deficiency
MO 2024	Most common cause of infantile diarrhoea worldwide	Rotavirus	Rotavirus vaccine is in India's UIP
MO 2024	Commonest heart defect in congenital rubella	PDA	Rubella triad: PDA, cataract, deafness
MO 2024	Bagel sign on ultrasound	Ectopic pregnancy	Tubal ring sign; 'snowstorm' = molar pregnancy
MO 2024	Graves' disease: thyroid profile	Low TSH	High T3/T4, suppressed TSH; TRAb positive

Paper	Question (shortened)	Answer	The fact to carry away
MO 2024	Cannot cross the placenta	Albumin and IgM	IgG is the only immunoglobulin that crosses
MO 2024	Not a structure on the right heart border on X-ray	Right ventricle	Right border = SVC, right atrium, IVC

Related items from GPSC's other medical papers: 'visible morphological change at cell level' = **atypia** (Public Health Dentist, 2024); perineural invasion = **adenoid cystic carcinoma** (same paper); the commonest primary metastasising to jaws = **breast** (same paper); ocular surface squamous neoplasia → test for **HIV** (Junior Ophthalmologist, 2023).

⚠ **One key to distrust:** the MO 2024 key gives Little's area as 'posterosuperior'. Every ENT text puts it in the **anteroinferior** septum. If it is asked again, mark anteroinferior.

Cell injury, adaptation and death

The single most-tested line: **irreversible injury = membrane damage + nuclear changes** (pyknosis → karyorrhexis → karyolysis). Swelling and fatty change are reversible.

Adaptations

Change	Meaning	MCQ example
Hypertrophy	Bigger cells	Heart in hypertension, uterus in pregnancy (also hyperplasia), skeletal muscle
Hyperplasia	More cells	Endometrium, BPH, breast at puberty; pathological hyperplasia can precede cancer
Atrophy	Smaller cells	Brain in old age, limb in plaster; ubiquitin–proteasome + autophagy
Metaplasia	One adult cell type replaced by another	Smoker's bronchus (columnar → squamous); Barrett oesophagus (squamous → intestinal columnar)
Dysplasia	Disordered growth: pleomorphism, high N:C ratio, loss of polarity	Cervix CIN; reversible until it crosses the basement membrane
Anaplasia	Lack of differentiation	Hallmark of malignancy

Necrosis types

Type	Where	Remember
Coagulative	Infarcts of heart, kidney, spleen (all solid organs except brain)	Ghost outlines of cells
Liquefactive	Brain infarct, abscess	Enzymes digest tissue
Caseous	Tuberculosis	Cheesy, amorphous, inside granulomas
Fat	Acute pancreatitis, breast trauma	Calcium soaps = saponification
Fibrinoid	Malignant hypertension, vasculitis	Bright pink vessel wall
Gangrenous	Limb	Dry = arterial; wet = venous + infection; gas = Clostridium perfringens

Apoptosis vs necrosis: apoptosis is programmed, single cells, **no inflammation**, cell shrinkage, chromatin condensation, apoptotic bodies, DNA ladder on electrophoresis, caspases. Intrinsic (mitochondrial, cytochrome c, BCL-2 family) and extrinsic (Fas–FasL, TNF) pathways. Example: Councilman bodies in viral hepatitis.

Free radicals and reperfusion: injury worsens when blood returns (reperfusion injury). Superoxide, H2O2, hydroxyl radical (most reactive); defences = SOD, catalase, glutathione peroxidase, vitamins E, C, A.

Intracellular accumulations and pigments

- Fatty change (steatosis): liver in alcohol, diabetes, obesity, kwashiorkor.
- **Lipofuscin** = 'wear-and-tear' brown pigment of ageing (heart, liver).
- **Haemosiderin** = iron; Prussian blue (Perls) stain; haemochromatosis.
- Melanin; bilirubin; anthracotic carbon in lungs.
- **Hyaline**: Russell bodies (plasma cells), Mallory–Denk bodies (alcoholic liver).

Calcification

Dystrophic	Metastatic
Normal serum calcium	High serum calcium

Dystrophic	Metastatic
In dead/damaged tissue: TB caseation, atheroma, damaged valves, psammoma bodies	In normal tissue: lungs, kidneys, gastric mucosa, blood vessels
Psammoma bodies: papillary thyroid Ca, serous ovarian Ca, meningioma, mesothelioma	Causes: hyperparathyroidism, vitamin D excess, bone destruction, renal failure

Stain for calcium: **von Kossa** or Alizarin red.

Memory hook: apoptosis is a **quiet suicide** (one cell, shrinks, no inflammation); necrosis is a **messy murder** (groups of cells, swelling, inflammation). The dying nucleus goes **P-R-L**: **P**yknosis shrinks it, karyo**R**rhexis breaks it, karyo**L**ysis dissolves it.

Inflammation and repair

Acute inflammation is neutrophil-led and lasts hours to days; chronic inflammation is led by macrophages, lymphocytes and plasma cells, with tissue destruction and fibrosis going on together.

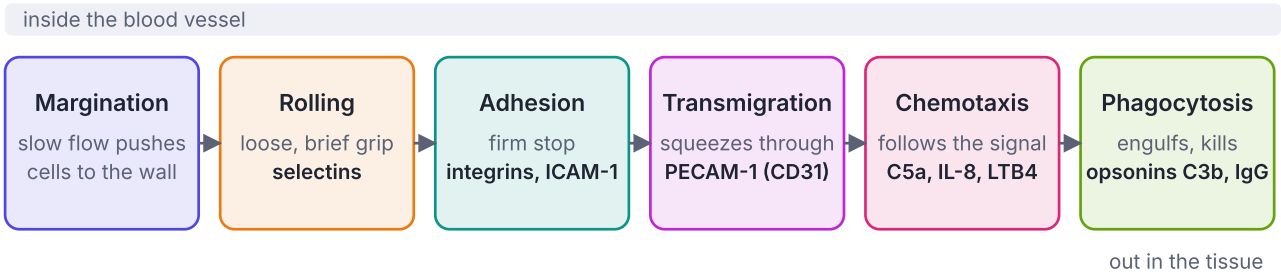
Cardinal signs: rubor, tumor, calor, dolor (Celsus); functio laesa added by Virchow.

Sequence of acute inflammation: transient vasoconstriction → vasodilatation → increased permeability (exudate) → stasis → leucocyte margination → rolling (**selectins**) → firm adhesion (**integrins** to ICAM-1/VCAM-1) → transmigration (**PECAM-1/CD31**, mainly in venules) → chemotaxis → phagocytosis → killing.

- Chemotactic agents: **C5a, LTB4, IL-8, bacterial products (fMLP)**.
- Opsonins: **IgG Fc and C3b** (also collectins).
- Most efficient microbicidal system: **H2O2–MPO–halide** (myeloperoxidase) system.
- Chronic granulomatous disease = defect in NADPH oxidase (no respiratory burst); NBT test negative.
- Chédiak–Higashi = defective phagolysosome fusion, giant granules.
- LAD type 1 = integrin (CD18) defect, delayed umbilical cord separation.

A white cell leaves the vessel in six steps

"My Royal Army Takes Charge", then eats; the molecule that drives each step is underneath



leucocyte exit cascade · 6 steps

Selectins make cells roll, integrins make them stop: MCQs swap these two.

Mediators

Mediator	Key action
Histamine (mast cells)	First mediator: vasodilatation and increased permeability (immediate transient response)
Serotonin	Like histamine; from platelets
Prostaglandins PGI2, PGE2	Vasodilatation; PGE2 = fever and pain
Thromboxane A2	Vasoconstriction, platelet aggregation
Leukotrienes C4, D4, E4	Bronchospasm, permeability (slow-reacting substance of anaphylaxis)
Bradykinin	Pain, permeability
C3a, C5a	Anaphylatoxins (histamine release); C5a also chemotactic
C5b–9	Membrane attack complex
IL-1, TNF, IL-6	Fever, acute-phase reactants, leucocytosis
NO	Vasodilatation; kills microbes

Aspirin and NSAIDs block **cyclo-oxygenase**; steroids block **phospholipase A2**.

Outcomes and patterns: resolution, abscess (pus), fibrosis, or chronic inflammation. Serous (blister, effusion), fibrinous (pericarditis — 'bread and butter'), suppurative (abscess, appendicitis), ulcer.

Granulomatous inflammation: a granuloma is a collection of **epithelioid cells** (activated macrophages) with or without Langhans giant cells, rimmed by lymphocytes.

Caseating	Non-caseating
Tuberculosis (classic); some fungi	Sarcoidosis, Crohn's disease, leprosy (tuberculoid), foreign body, berylliosis

Giant cells: **Langhans** = nuclei at the periphery in a horseshoe (TB); **foreign body** = scattered nuclei; **Touton** = xanthoma; **Reed–Sternberg** = Hodgkin lymphoma; **Warthin–Finkeldey** = measles; **Aschoff bodies / Anitschkow cells** = rheumatic carditis.

Systemic effects: fever (IL-1, TNF, PGE2 on the hypothalamus), acute-phase proteins (CRP, fibrinogen, serum amyloid A), raised ESR, leucocytosis. Neutrophilia = bacteria; lymphocytosis = viruses, TB, pertussis; eosinophilia = allergy, parasites; leucopenia = typhoid, dengue, some viral infections.

Repair

- Labile cells (skin, mucosa, bone marrow) always divide; stable cells (liver, kidney) divide when needed; **permanent cells (neurons, cardiac muscle)** cannot — they heal by scar.
- **Granulation tissue** = new capillaries (angiogenesis, VEGF) + fibroblasts + inflammatory cells; it is the hallmark of healing.
- Myofibroblasts cause **wound contraction**.
- First intention (clean, sutured): epithelium covers in 24–48 h; collagen type III early, replaced by type I; strength ~70–80% of normal by 3 months.
- Second intention: larger gap, more granulation tissue, contraction.
- Delayed healing: infection (commonest), poor blood supply, diabetes, steroids, vitamin C deficiency (hydroxylation of proline/lysine), zinc deficiency, foreign body.
- **Keloid** extends beyond the wound margin; **hypertrophic scar** stays within it.

Memory hook: the four Latin signs **rubor, tumor, calor, dolor** come from Celsus; Virchow added the fifth, **functio laesa**. White cells leave the vessel as "**M**y **R**oyal **A**rmy **T**akes **C**harge": **M**argination, **R**olling (selectins), **A**dhesion (integrins), **T**ransmigration (PECAM-1), **C**hemotaxis.

Haemodynamics: oedema, thrombosis, embolism, infarction, shock

The three classics here are Virchow's triad, the commonest source of pulmonary embolism (deep veins of the legs) and the organs that get red versus pale infarcts.

Oedema

Transudate	Exudate
Protein < 3 g/dL, specific gravity < 1.012	Protein > 3 g/dL, specific gravity > 1.020
Few cells	Many cells (inflammatory)
Raised hydrostatic pressure (CHF), low oncotic pressure (nephrotic syndrome, cirrhosis)	Inflammation, increased permeability

Light's criteria for an exudative pleural effusion: fluid/serum protein > 0.5, fluid/serum LDH > 0.6, fluid LDH > two-thirds of the upper normal serum value.

Thrombosis

- **Virchow's triad**: endothelial injury (most important, alone can cause thrombosis), abnormal flow (stasis or turbulence), hypercoagulability.
- **Lines of Zahn** = alternating pale platelet–fibrin and dark red-cell layers; they prove a thrombus formed in flowing blood (ante-mortem). Post-mortem clots are rubbery, 'chicken-fat' over 'currant jelly', and not attached.
- Commonest inherited hypercoagulable state: **Factor V Leiden** (resistance to activated protein C). Acquired: antiphospholipid syndrome, cancer (Trousseau migratory thrombophlebitis), pregnancy, OCPs, immobilisation, surgery.
- Fate: propagation, embolisation, dissolution, organisation and recanalisation.

Embolism

Type	Key facts
Pulmonary thromboembolism	Commonest embolism; > 95% from deep leg veins (popliteal and above); saddle embolus at the bifurcation can kill instantly
Systemic	Mostly from heart (mural thrombus after MI, AF, valve vegetations); lodge in lower limbs > brain
Fat	Long-bone fractures; 1–3 days later: breathlessness, confusion, petechial rash; fat globules stained by Oil Red O / Sudan
Air	> 100 mL needed; decompression sickness (nitrogen) = caisson disease, 'bends', 'chokes'

Type	Key facts
Amniotic fluid	Labour; squames and lanugo hair in pulmonary vessels; DIC

Infarction

White (pale, anaemic)	Red (haemorrhagic)
Arterial occlusion in solid organs with end-arteries: heart, kidney, spleen	Venous occlusion (testicular/ovarian torsion), loose tissue, dual blood supply: **lung, liver**, small intestine; reperfusion

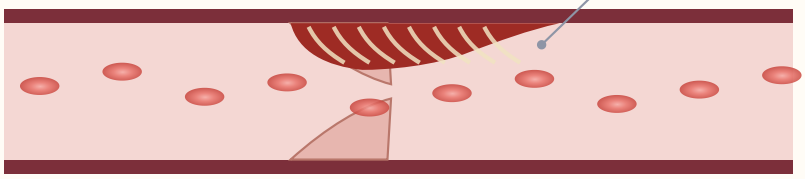
Infarcts are wedge-shaped with the apex at the occluded vessel. Brain infarcts liquefy; others undergo coagulative necrosis. Neurons die after 3–4 minutes of ischaemia, myocardium after 20–30 minutes.

Thrombosis, embolism and infarction

Why clots form, where they travel, what they kill

VIRCHOW'S TRIAD

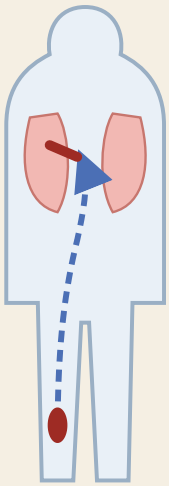
I — Endothelial injury
Most important: alone can cause thrombosis



S — Stasis or turbulence
Abnormal flow: blood pooling or swirling

H — Hypercoagulability
Factor V Leiden (inherited); pregnancy, OCPs, cancer, immobilisation, surgery

Embolus: a clot that travels

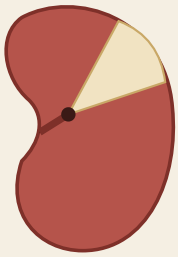


Pulmonary embolism: commonest embolism; > 95% from deep leg veins (popliteal and above)

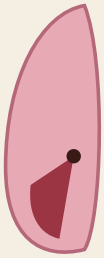
Saddle embolus at the bifurcation can kill instantly

Others: fat (long-bone fractures), air, amniotic fluid

Infarct: wedge-shaped, apex at the block



White (pale) infarct
arterial block in solid organs with end-arteries: heart, kidney, spleen



Red infarct
venous block, loose tissue or dual supply: lung, liver, small intestine

Thrombosis to infarct · Virchow's triad, the travelling clot, pale vs red infarcts

Myocardial infarction timeline (histology): up to 4 h nothing visible; 4–24 h coagulative necrosis, wavy fibres; 1–3 days neutrophils; 3–7 days macrophages (rupture risk highest); 7–10 days granulation tissue; 2 months scar.

Shock

Type	Mechanism	Example
Hypovolaemic	Low blood volume	Haemorrhage, burns, diarrhoea
Cardiogenic	Pump failure	MI, arrhythmia, tamponade
Septic (distributive)	Vasodilatation from bacterial products (endotoxin = LPS of Gram-negatives)	Gram-negative sepsis
Neurogenic / anaphylactic	Loss of vascular tone	Spinal injury; IgE-mediated

Stages: non-progressive (compensated) → progressive (lactic acidosis) → irreversible. Morphology: 'shock lung' (diffuse alveolar damage / ARDS), acute tubular necrosis, adrenal lipid depletion; Waterhouse–Friderichsen syndrome (adrenal haemorrhage) in meningococcal sepsis.

DIC: widespread microthrombi + bleeding; low platelets, prolonged PT and aPTT, low fibrinogen, **raised D-dimer / FDP**, schistocytes on smear. Causes: sepsis, obstetric (abruption, amniotic fluid embolism, dead fetus), malignancy (acute promyelocytic leukaemia), snakebite (viper).

Memory hook: Virchow's triad is **I-S-H**: endothelial **I**njury, **S**tasis, **H**ypercoagulability. A thrombus forms in a living vessel, a clot can form anywhere, and an embolus is anything that travels and lodges.

Neoplasia

The single feature that defines malignancy is **invasion/metastasis**; anaplasia is its microscopic hallmark. Carcinomas spread by lymphatics, sarcomas by blood.

Naming traps

Sounds benign, is malignant	Is benign despite the name
Lymphoma, melanoma, seminoma, mesothelioma, hepatoma	Hamartoma (disorganised normal tissue), choristoma (normal tissue in wrong place, e.g. pancreatic rest)

Teratoma: all three germ layers; mature cystic teratoma (dermoid) of ovary is benign.

Benign vs malignant: differentiation, rate of growth, local invasion (capsule), metastasis. **Carcinoma in situ** = full-thickness dysplasia, basement membrane intact. Basal cell carcinoma and gliomas rarely metastasise.

Spread

- Lymphatic: carcinomas; first node = sentinel node; **Virchow's (Troisier's) node** = left supraclavicular, gastric carcinoma.
- Haematogenous: sarcomas, renal cell carcinoma (renal vein), hepatocellular carcinoma (hepatic vein), follicular thyroid carcinoma, choriocarcinoma. Liver and lungs are the commonest sites.
- Transcoelomic: ovary; **Krukenberg tumour** = signet-ring metastasis to ovaries from stomach; Sister Mary Joseph nodule = umbilical metastasis.
- Bone metastases: osteolytic mostly; **osteoblastic in prostate** (and breast sometimes).

Carcinogens

Agent	Cancer
Tobacco (polycyclic hydrocarbons, nitrosamines)	Lung, oral, larynx, oesophagus, bladder, pancreas
Aflatoxin B1 (Aspergillus flavus)	Hepatocellular carcinoma (p53 mutation)
Asbestos	Mesothelioma (and lung, with smoking)
Vinyl chloride	Angiosarcoma of liver
Aromatic amines, aniline dyes, Schistosoma haematobium	Bladder (Schistosoma → squamous)
Benzene	Acute myeloid leukaemia
Ionising radiation	Leukaemia, papillary thyroid carcinoma
UV light (UVB)	Skin: BCC, SCC, melanoma; xeroderma pigmentosum

Oncogenic microbes

Microbe	Cancer
HPV 16, 18	Cervix, anal, oropharyngeal carcinoma (E6 → p53; E7 → RB)

Microbe	Cancer
EBV	Burkitt lymphoma, nasopharyngeal carcinoma, Hodgkin lymphoma
HBV, HCV	Hepatocellular carcinoma
HTLV-1	Adult T-cell leukaemia/lymphoma
HHV-8	Kaposi sarcoma
Helicobacter pylori	Gastric adenocarcinoma, MALT lymphoma

Genes: oncogenes need one hit (RAS — commonest mutated oncogene, MYC, ERBB2/HER2, BCR–ABL in CML from t(9;22) Philadelphia chromosome). Tumour suppressors need two hits (Knudson): **RB** (retinoblastoma, osteosarcoma), **p53** (guardian of the genome; commonest mutated gene in human cancer; Li–Fraumeni), APC (colon), BRCA1/2 (breast, ovary), WT1 (Wilms), NF1.

Tumour markers

Marker	Cancer
AFP	Hepatocellular carcinoma, yolk sac tumour
β-hCG	Choriocarcinoma, molar pregnancy, testicular tumours
CEA	Colorectal carcinoma (follow-up)
CA-125	Ovarian (epithelial) carcinoma
CA 19-9	Pancreatic carcinoma
CA 15-3	Breast carcinoma
PSA	Prostate carcinoma
Calcitonin	Medullary thyroid carcinoma
Thyroglobulin	Follow-up of papillary/follicular thyroid cancer
LDH, β2-microglobulin	Lymphoma, myeloma

Immunohistochemistry: **cytokeratin** = carcinoma; **vimentin** = sarcoma; **desmin** = muscle; **S-100 / HMB-45** = melanoma; **LCA (CD45)** = lymphoma; **GFAP** = glial tumours; **chromogranin, synaptophysin** = neuroendocrine.

Grading and staging: grade = degree of differentiation (microscopic); **stage = size + spread (TNM)** and is the better predictor of prognosis.

Paraneoplastic syndromes: SIADH and Cushing's (small-cell lung carcinoma); hypercalcaemia via PTHrP (squamous-cell lung carcinoma); polycythaemia (renal cell carcinoma, hepatoma); Lambert–Eaton (small-cell); acanthosis nigricans (gastric cancer); Trousseau sign (pancreas).

Commonest cancers in Indian men: lip and oral cavity, lung. **In women**: breast, then cervix. Pap smear screens for cervical cancer; VIA is the low-cost screening tool used in India.

Memory hook: **grade** is how ugly the cells look; **stage** is how far the tumour has travelled, and stage predicts the outcome better. For stains: **C**ytokeratin for **C**arcinoma, **V**imentin for sarcoma ("**V**ery soft tissue"), **D**esmin for muscle, **S**-100 for melanoma.

Immunity, hypersensitivity and amyloidosis

The hypersensitivity table answers most questions here; learn one example per type.

Immunoglobulins

Class	Remember
IgG	Most abundant in serum (~75%); only one to cross the placenta; secondary response; 4 subclasses
IgA	Secretions (milk, saliva, tears, gut); dimer with secretory component; colostrum
IgM	Largest (pentamer); first in primary response; best complement activator; IgM in serum = recent infection; does not cross placenta (fetal IgM = congenital infection)
IgE	Lowest level; binds mast cells; allergy and helminths; heat-labile
IgD	B-cell surface receptor

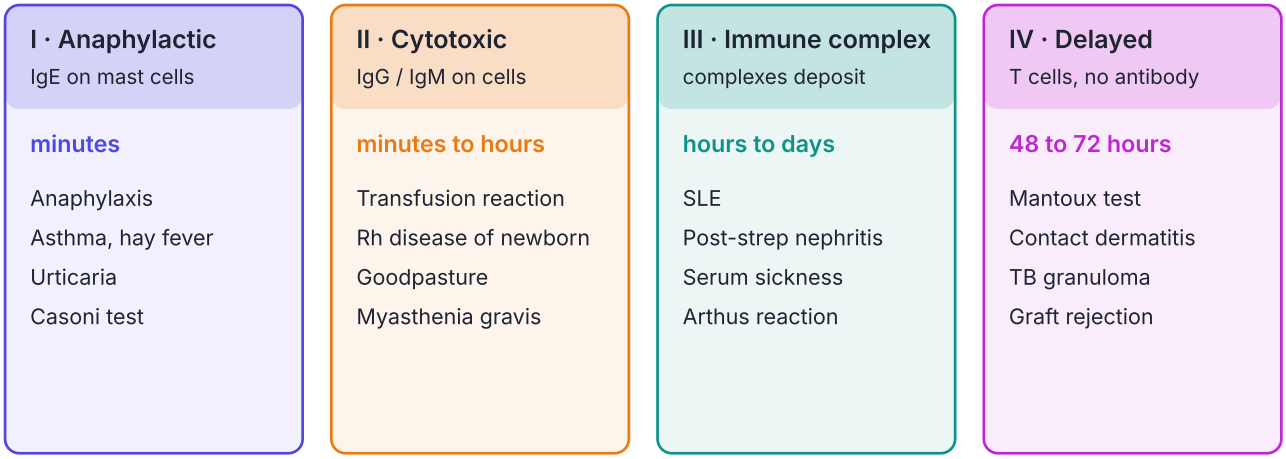
T cells mature in the **thymus**; B cells in the bone marrow. CD4 = helper (MHC II); CD8 = cytotoxic (MHC I). HIV binds CD4 with CCR5/CXCR4 co-receptors. NK cells kill without prior sensitisation (CD16, CD56).

Hypersensitivity (Gell and Coombs)

Type	Mechanism	Examples
I — immediate (anaphylactic)	IgE on mast cells → histamine	Anaphylaxis, asthma, hay fever, urticaria; Casoni test (old)
II — cytotoxic	IgG/IgM against cell or tissue antigens	Transfusion reaction, haemolytic disease of newborn, Goodpasture, Graves' (stimulating), myasthenia gravis, autoimmune haemolytic anaemia, rheumatic fever
III — immune complex	Antigen–antibody complexes deposit, activate complement	SLE, post-streptococcal glomerulonephritis, serum sickness, Arthus reaction, polyarteritis nodosa
IV — delayed (cell-mediated)	Sensitised T cells, 48–72 h	Tuberculin (Mantoux) test, contact dermatitis, granulomas, transplant rejection, lepromin test

Hypersensitivity spells ACID: four types, four speeds

What drives each type, how fast it appears, and the examples asked most



Only type IV involves no antibody; the Mantoux test is read at 48 to 72 hours for that reason.

hypersensitivity types I to IV · ACID

Read the speed row first: minutes, hours, days, then two to three days for the T-cell type.

Vaccines (types): live attenuated = BCG, OPV, MMR, rotavirus, yellow fever, chickenpox; killed/inactivated = IPV, hepatitis A, rabies, pertussis (whole cell); toxoid = tetanus, diphtheria; subunit/recombinant = hepatitis B, HPV. Live vaccines are avoided in pregnancy and immunosuppression.

Autoantibodies worth memorising

Antibody	Disease
Anti-dsDNA, anti-Smith	SLE (most specific)

Antibody	Disease
ANA	SLE screening (most sensitive)
Anti-histone	Drug-induced lupus
Anti-CCP, rheumatoid factor	Rheumatoid arthritis (anti-CCP more specific)
Anti-centromere	Limited systemic sclerosis (CREST)
Anti-Scl-70 (topoisomerase)	Diffuse systemic sclerosis
Anti-Ro, anti-La	Sjögren syndrome; congenital heart block
c-ANCA (PR3) / p-ANCA (MPO)	Granulomatosis with polyangiitis / microscopic polyangiitis
Anti-TPO	Hashimoto thyroiditis
TSH-receptor antibody (TRAb)	Graves' disease
Anti-tTG IgA	Coeliac disease
Anti-AChR	Myasthenia gravis

Amyloidosis: extracellular deposition of misfolded fibrillar protein in a **β-pleated sheet**.

- Stains: **Congo red** — pink-red, and **apple-green birefringence under polarised light** (the diagnostic test). Also crystal violet/methyl violet (metachromasia), thioflavin T.
- Types: **AL** (light chains: multiple myeloma, primary); **AA** (serum amyloid A: chronic inflammation — TB, rheumatoid arthritis, bronchiectasis, osteomyelitis, leprosy); Aβ2-microglobulin (long-term haemodialysis); Aβ (Alzheimer's); ATTR (senile cardiac, familial); calcitonin (medullary thyroid carcinoma).
- Biopsy sites: **abdominal fat pad aspiration** or rectal biopsy (least invasive); kidney is the commonest organ affected (nephrotic syndrome).
- Spleen: 'sago spleen' (follicular deposits) and 'lardaceous spleen' (red pulp/sinusoidal).

Transplant rejection: hyperacute (minutes; preformed antibodies), acute (weeks–months; T cells), chronic (months–years; vascular fibrosis). Graft-versus-host disease follows bone-marrow transplant (skin, liver, gut).

Memory hook: hypersensitivity is **ACID**: Type I **A**naphylactic (IgE), II **C**ytotoxic (antibody on cells), III **I**mmune complex, IV **D**elayed (T cells). For antibodies: Ig**M** is **M**ade first, Ig**G** alone **G**oes across the placenta, Ig**A** guards the secretions, Ig**E** causes allergy.

General microbiology: sterilisation, stains, media, tests

Three pairings carry most marks here: the sterilising method for each material, the stain for each organism and the medium for each organism.

Sterilisation and disinfection

Method	Conditions	Used for
Autoclave (moist heat)	**121 °C, 15 psi (1.05 kg/cm²), 15–20 min**	Surgical instruments, dressings, culture media; best method. Test organism: **Geobacillus stearothermophilus** spores
Hot-air oven (dry heat)	**160 °C for 2 h** (or 180 °C, 30 min)	Glassware, oils, powders, sharp instruments. Test: Bacillus atrophaeus (subtilis) spores
Pasteurisation	63 °C 30 min (holder) or 72 °C 15–20 s (flash)	Milk; kills Brucella, Mycobacterium bovis, Coxiella (most resistant of milk pathogens)
Tyndallisation	100 °C, 20 min on 3 successive days	Sugar and egg media
Inspissation	80–85 °C, 30 min, 3 days	Löwenstein–Jensen and Löffler serum slope
Incineration	—	Anatomical waste, soiled dressings
Ethylene oxide	Gas	Heat-sensitive items: plastics, heart-lung machines, catheters
Glutaraldehyde 2% (Cidex)	Liquid	Endoscopes, cystoscopes
Ionising (gamma) radiation	Cold sterilisation	Disposable syringes, catheters, sutures
Filtration (0.22 µm membrane)	—	Sera, antibiotic and sugar solutions

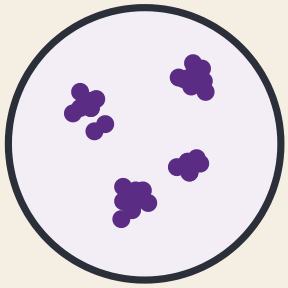
Prions resist routine sterilisation (need autoclaving at 134 °C for longer, plus NaOH). Bacterial **spores** are the most resistant living form; endospores are found in Bacillus and Clostridium.

Stains

Stain	Detects
Gram	Gram-positive (purple, thick peptidoglycan) vs Gram-negative (pink, LPS outer membrane). Steps: crystal violet → Gram's iodine (mordant) → decolourise (acetone/alcohol) → safranin
Ziehl–Neelsen (acid-fast)	M. tuberculosis (20% H2SO4), M. leprae (5% H2SO4); Nocardia (1%); Cryptosporidium, Isospora oocysts (modified)
Albert / Neisser	Metachromatic (volutin) granules of C. diphtheriae — 'Chinese letter' arrangement
India ink (negative stain)	Capsule of Cryptococcus neoformans in CSF
Giemsa / Leishman / JSB	Malaria parasites, blood films; Giemsa also Chlamydia inclusions, Donovan bodies
KOH mount (10%)	Fungal hyphae in skin, hair, nails
Lactophenol cotton blue	Fungal morphology in culture
Silver (Levaditi, Fontana)	Spirochaetes: Treponema, Leptospira
Dark-ground microscopy	Live Treponema pallidum, Leptospira
Auramine–rhodamine (fluorescent)	AFB screening — faster than ZN

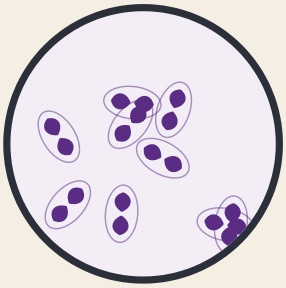
Down the microscope

Shape, arrangement and stain identify the organism



Gram-positive cocci in clusters

Staphylococcus aureus



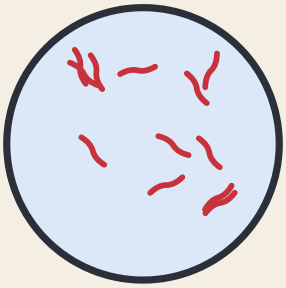
Lancet-shaped diplococci

Streptococcus pneumoniae



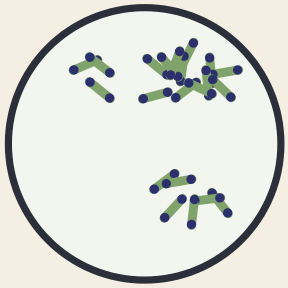
Gram-negative rods

E. coli
lactose fermenter



Acid-fast bacilli (Ziehl–Neelsen)

M. tuberculosis
red rods on blue



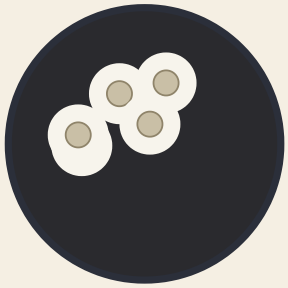
Albert stain: "Chinese letters"

C. diphtheriae,
volutin granules



Spirochaetes (dark ground)

Treponema pallidum,
Leptospira



India ink: clear capsule

Cryptococcus neoformans in CSF

Gram stain: 4 steps

- Crystal violet
- Gram's iodine
- Decolourise
- Safranin

Positive stays purple (thick wall); negative turns pink (LPS)

Down the microscope · shape, arrangement and stain name the organism

Culture media

Medium	Organism / type
Löwenstein–Jensen (LJ)	M. tuberculosis (6–8 weeks); malachite green inhibits contaminants
Löffler serum slope; potassium tellurite	C. diphtheriae
TCBS, alkaline peptone water (enrichment)	Vibrio cholerae (yellow colonies)
MacConkey	Gram-negative enterics; lactose fermenters pink (E. coli, Klebsiella)

Medium	Organism / type
Blood agar	Haemolysis: β (complete, Strep. pyogenes), α (green, Strep. pneumoniae, viridans), γ (none)
Chocolate agar	Haemophilus (needs X and V factors), Neisseria
Thayer–Martin	Neisseria gonorrhoeae
Wilson–Blair, DCA, XLD, SS agar	Salmonella / Shigella
Bile broth / blood culture	Salmonella Typhi (first week)
EMJH, Korthof's	Leptospira
Sabouraud dextrose agar	Fungi
Robertson's cooked meat	Anaerobes (Clostridium)
Bordet–Gengou	Bordetella pertussis ('bisected pearl' colonies)
NNN	Leishmania

Endotoxin = lipid A of the LPS in Gram-negative walls (fever, septic shock); Gram-positives lack it. **Exotoxins** = proteins, mostly Gram-positive, toxoids possible (tetanus, diphtheria, botulinum, cholera). Most potent toxin: **botulinum**.

Serological and lab tests

Test	Principle / use
ELISA	Screening (HIV, hepatitis, dengue NS1) — high sensitivity
Western blot	Confirmation of HIV (older algorithm)
PCR / RT-PCR	DNA / RNA detection; RT-PCR for COVID-19, HIV viral load
CBNAAT (GeneXpert) / TrueNat	TB + rifampicin resistance in ~2 h
Widal	Tube agglutination, typhoid (O and H antibodies); rising titre matters
VDRL / RPR	Non-treponemal flocculation test for syphilis; false positives (SLE, pregnancy, leprosy)
TPHA / FTA-ABS	Treponemal, confirmatory for syphilis

Test	Principle / use
ASO titre	Recent streptococcal infection (rheumatic fever)
Weil–Felix	Heterophile agglutination with Proteus strains for rickettsiae (scrub typhus = OX-K)
Paul–Bunnell / Monospot	Infectious mononucleosis (EBV)
Coombs (direct / indirect)	Antibody on RBCs / in serum: haemolytic anaemia, Rh disease
Mantoux (tuberculin, 5 TU PPD)	Type IV; read at 48–72 h; ≥ 10 mm positive
Schick test (historic)	Immunity to diphtheria
Lepromin	Prognosis, not diagnosis, of leprosy (Mitsuda reaction at 3–4 weeks)

Sensitivity vs specificity: sensitivity = true positives found among the diseased (screening tests need high sensitivity); specificity = true negatives among the healthy (confirmatory tests need high specificity).

Memory hook: the autoclave rule is **121, 15, 15**: 121 °C at 15 lb per square inch for 15 minutes kills spores. Prions shrug off even that, which is why they are the hardest of all to sterilise.

Bacteria

Tuberculosis and typhoid are asked most; leptospirosis and scrub typhus are the Goa monsoon specials.

Tuberculosis (Mycobacterium tuberculosis)

- Acid-fast, slow growing (generation time ~18 h), obligate aerobe; cord factor (virulence); grows on LJ in 6–8 weeks, faster in liquid MGIT culture.
- **Primary TB**: Ghon focus (subpleural, lower part of upper lobe / upper part of lower lobe) + hilar node = **Ghon complex**; healed and calcified = Ranke complex.
- **Secondary (post-primary)**: apex of the lung (high oxygen) — **Simon focus** = apical seeding; **Assmann focus** = infraclavicular; cavitation; Rasmussen aneurysm causes haemoptysis.
- Miliary TB: millet-seed lesions; spine = **Pott's disease** (thoracolumbar, cold abscess, gibbus); TB of the bone: best diagnosis = biopsy (MO 2025).

- Diagnosis under NTEP: sputum smear for AFB (two samples) and **NAAT (CBNAAT/TrueNat) as initial test**; culture is the gold standard. Mantoux and IGRA show infection, not disease. ADA raised in TB pleural fluid.
- MDR-TB = resistant to isoniazid + rifampicin; XDR adds fluoroquinolone + one Group A drug (bedaquiline/linezolid).

Leprosy (M. leprae): cannot be grown in culture (grown in mouse footpad, nine-banded armadillo); prefers cool parts and nerves (Schwann cells); slit-skin smear, ZN with 5% H_2SO_4; 'globi' in lepromatous; lepromin test negative in lepromatous, positive in tuberculoid.

Enteric fever (Salmonella Typhi, Paratyphi A and B)

- Faeco-oral; incubation 10–14 days; step-ladder fever, relative bradycardia, rose spots, splenomegaly, **leucopenia**.
- Pathology: Peyer's patches of terminal ileum; ulcers are **longitudinal (oval, along the long axis of gut)** — perforation in week 3. (TB ulcers are transverse/girdle.)
- Investigation by week (**BASU**): week 1 **B**lood culture (also bone-marrow culture — most sensitive overall), week 2 **A**gglutination (Widal), week 3 **S**tool, week 4 **U**rine. Carrier site: gall bladder.

Other key bacteria

Organism	Disease and key fact
Vibrio cholerae O1 (El Tor) and O139	Rice-water stools; toxin raises cAMP; darting motility; TCBS; treatment = ORS
Corynebacterium diphtheriae	Grey pseudomembrane that bleeds on removal; bull neck; toxin inhibits EF-2; Albert stain; Elek's gel precipitation test for toxin
Streptococcus pyogenes (group A)	Pharyngitis → rheumatic fever (ASO), post-streptococcal glomerulonephritis; bacitracin-sensitive
Streptococcus pneumoniae	Lobar pneumonia; lancet-shaped diplococci; optochin-sensitive; bile soluble; Quellung
Staphylococcus aureus	Coagulase-positive; boils, osteomyelitis, toxic shock, food poisoning (preformed toxin, 1–6 h); MRSA
Clostridium tetani	Drumstick spores; tetanospasmin blocks inhibitory transmitters (GABA, glycine); risus sardonicus

Organism	Disease and key fact
Clostridium perfringens	Gas gangrene; Nagler reaction (alpha toxin = lecithinase)
Clostridium botulinum	Blocks acetylcholine release; descending flaccid paralysis; canned food
Neisseria meningitidis	Meningitis with petechial rash; Waterhouse–Friderichsen
Treponema pallidum	Syphilis: painless chancre (primary), condylomata lata (secondary), gumma (tertiary); dark-ground microscopy; VDRL then TPHA
Helicobacter pylori	Peptic ulcer, gastric cancer; urease breath test; Warthin–Starry silver stain
Bordetella pertussis	Whooping cough; lymphocytosis; cough-plate culture

Leptospirosis (Leptospira interrogans) — common in Goa after monsoon paddy work.

- Spread through **rat urine** in water/soil entering through abraded skin or mucosa.
- Biphasic illness; **conjunctival suffusion**, calf tenderness; severe form = **Weil's disease** (jaundice, renal failure, bleeding); pulmonary haemorrhage.
- Diagnosis: first week blood/CSF culture (EMJH) or PCR; after day 7 **IgM ELISA**; MAT (microscopic agglutination test) = reference; dark-ground microscopy. Treatment doxycycline/penicillin; doxycycline chemoprophylaxis for high-risk exposure.

Scrub typhus (Orientia tsutsugamushi) — transmitted by the **larval trombiculid mite (chigger)**; painless **eschar** at the bite site; fever, lymphadenopathy, pneumonitis. Diagnosis IgM ELISA (Weil–Felix OX-K is old). Drug: **doxycycline** (azithromycin in pregnancy and children).

Other rickettsiae: epidemic typhus (louse, R. prowazekii), endemic typhus (flea), Indian tick typhus (tick, R. conorii), Q fever (Coxiella, inhalation, no rash).

Memory hook: typhoid tests follow the weeks as **BASU**: **B**lood culture in week 1, **A**ntibody (Widal) in week 2, **S**tool culture in week 3, **U**rine in week 4. TB is **red rods on blue** with Ziehl–Neelsen.

Viruses

Hepatitis serology and dengue lab tests are the high-yield core; know which marker means what.

Hepatitis viruses

Virus	Genome / family	Spread	Remember
HAV	RNA, picornavirus	Faeco-oral	No carrier state; anti-HAV IgM = acute
HBV	**DNA** (partly double-stranded), hepadnavirus	Blood, sexual, vertical	Carrier state, chronic hepatitis, cirrhosis, HCC; vaccine available
HCV	RNA, flavivirus	Blood (transfusion, IV drug use)	Most chronicity (~80%); no vaccine; direct-acting antivirals cure
HDV	Defective RNA	Needs HBsAg	Co-infection or superinfection with HBV
HEV	RNA	Faeco-oral, water-borne epidemics	**High mortality in pregnant women (~20%)**

Hepatitis B serology

Marker	Meaning
HBsAg	First marker; infection (acute or chronic); > 6 months = chronic carrier
HBeAg	Active replication, high infectivity
Anti-HBc IgM	Acute (recent) infection; the only marker in the 'window period'
Anti-HBc IgG	Past or chronic infection
Anti-HBs	Immunity (after recovery or vaccination); vaccinated persons have anti-HBs only
HBV DNA	Replication; guides treatment

Dengue (flavivirus, 4 serotypes) — vector **Aedes aegypti** (day-biter, breeds in clean stored water).

- Day 1–5: **NS1 antigen** or RT-PCR. After day 5: **IgM ELISA**. Secondary infection with another serotype risks severe dengue (antibody-dependent enhancement).
- Labs: **thrombocytopenia**, leucopenia, **rising haematocrit (≥ 20% = plasma leakage)**; positive tourniquet test (≥ 10–20 petechiae per square inch).
- Warning signs: abdominal pain, persistent vomiting, mucosal bleed, lethargy, liver > 2 cm, rising haematocrit with falling platelets.
- Chikungunya (alphavirus, also Aedes): severe, long-lasting arthralgia.

- Japanese encephalitis: Culex; pigs = amplifier hosts, ardeid birds = reservoir; JE vaccine in UIP for endemic districts.
- **Kyasanur Forest Disease** (flavivirus, tick Haemaphysalis spinigera; monkeys die first): reported in Goa's Sattari taluka and Karnataka's Western Ghats.

- **HIV**: retrovirus (RNA → DNA via **reverse transcriptase**); gp120 binds CD4, co-receptors CCR5/CXCR4.
- Window period ~3–12 weeks (shorter with 4th-generation antigen–antibody tests).
- Indian testing strategy: three different rapid/ELISA tests for diagnosis (no Western blot needed).
- AIDS = CD4 < 200/µL or an AIDS-defining illness.
- Monitoring: CD4 count and **viral load**.
- Infant diagnosis < 18 months = DNA PCR (maternal antibodies confuse ELISA).

Other viruses to recognise

Virus	Key fact
Rabies (rhabdovirus, bullet-shaped)	**Negri bodies** in hippocampus and Purkinje cells; corneal impression smear / skin biopsy for antigen
Measles (paramyxovirus)	Koplik spots; Warthin–Finkeldey giant cells; SSPE late
Mumps	Parotitis, orchitis; aseptic meningitis
Rubella	Congenital rubella: PDA, cataract, deafness (MO 2024)
Varicella-zoster	Tzanck smear: multinucleated giant cells (also HSV); zoster = reactivation in dorsal root ganglion
EBV	Infectious mononucleosis, atypical lymphocytes (Downey cells), Monospot
Rotavirus	Commonest cause of infantile diarrhoea (MO 2024); wheel-shaped
Poliovirus	Faeco-oral; anterior horn cells; OPV + IPV
Influenza	Antigenic shift (pandemics, influenza A) vs drift (epidemics)
SARS-CoV-2	NP swab RT-PCR (MO 2025); ACE-2 receptor
HPV	Koilocytes in Pap smear; types 16, 18 cancer; 6, 11 warts

Inclusion bodies: **Negri** (rabies), **Guarnieri** (vaccinia/smallpox), **Cowdry type A** (herpes), **owl's eye** (CMV), **Molluscum bodies** (molluscum contagiosum).

Memory hook: "**The vowels hit the bowels**": hepatitis **A** and **E** spread by the faecal–oral route; **B, C and D** by blood and sex. Hepatitis D needs B to exist. Negri bodies mean **rabies**.

Parasites and fungi

Malaria carries the most marks: the species differences, the smear, and which stage is infective to whom.

Malaria (Plasmodium)

- Vector: female **Anopheles** (in Goa, urban malaria vector *An. stephensi* breeds in construction-site water). Man = intermediate host; mosquito = definitive host.
- Infective to man: **sporozoite** (mosquito bite). Infective to mosquito: **gametocyte**. Hypnozoites (dormant liver stage) in **vivax and ovale** cause relapse → primaquine.
- **Gold standard: peripheral smear** — thick smear to detect, thin smear to identify species; Giemsa/JSB/Leishman. RDT: HRP-2 = falciparum; pLDH = pan-species.

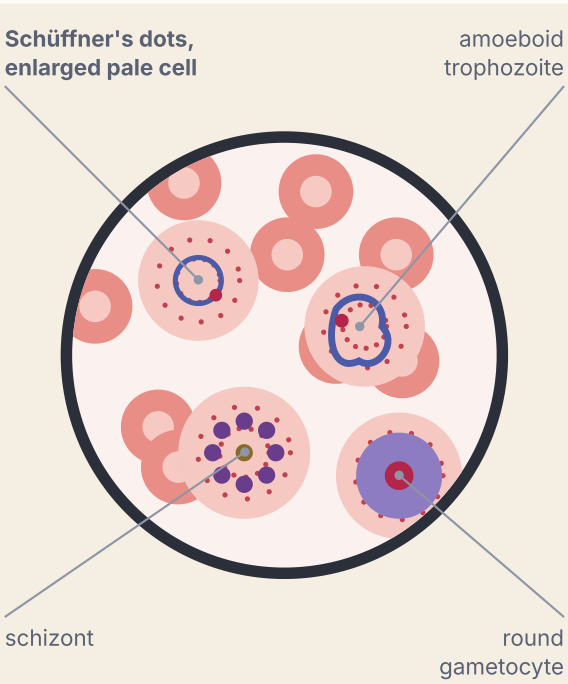
Feature	P. vivax	P. falciparum
RBC size	Enlarged, pale	Normal
Dots	Schüffner's dots	Maurer's clefts
Ring forms	Single, in one RBC	Multiple rings per RBC, accolé forms, double chromatin
Gametocyte	Round	**Crescent / banana shaped** (diagnostic)
Schizonts in smear	Seen	Usually not (sequestered in capillaries)
Fever cycle	Tertian (48 h)	Malignant tertian; cerebral malaria, blackwater fever

P. malariae = quartan (72 h), band forms, nephrotic syndrome. P. knowlesi = 24 h (monkey malaria, SE Asia).

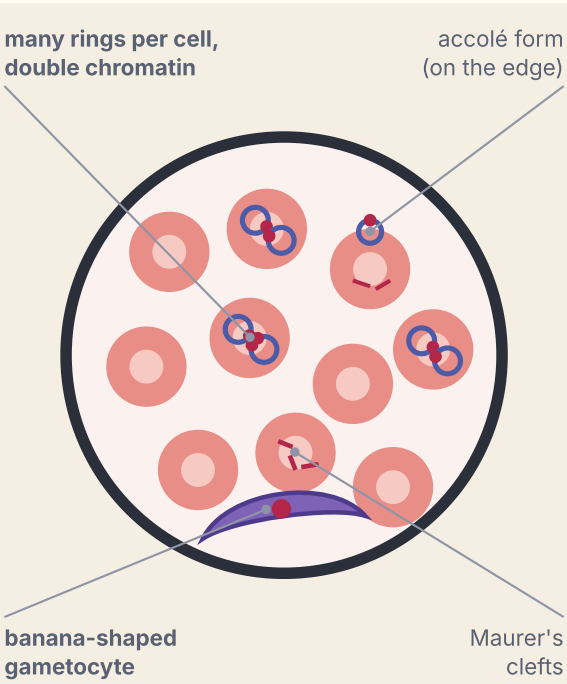
Malaria smear: vivax vs falciparum

Thin film, Giemsa stain

Plasmodium vivax



Plasmodium falciparum



	Plasmodium vivax	Plasmodium falciparum
RBC size	Enlarged, pale	Normal
Dots	Schüffner's dots	Maurer's clefts
Rings	Single, one per RBC	Many per RBC, accolé, double chromatin
Gametocyte	Round	Crescent / banana (diagnostic)
Schizonts in smear	Seen	Usually not (sequestered)
Fever	Tertian, 48 h	Malignant tertian; cerebral malaria

Relapse: vivax and ovale hide in the liver as hypnozoites → primaquine.
Infective to man: sporozoite (mosquito bite). Infective to mosquito: gametocyte.
Thick smear detects, thin smear identifies the species (Giemsa, JSB, Leishman).

Malaria smear · vivax vs falciparum on a thin film

Protozoa

Parasite	Disease and lab fact
Entamoeba histolytica	Amoebic dysentery (flask-shaped ulcers), **amoebic liver abscess ('anchovy sauce' pus)**, right lobe; trophozoite with ingested RBCs; cyst with 4 nuclei

Parasite	Disease and lab fact
Giardia lamblia	Malabsorption, steatorrhoea; 'falling-leaf' motility, pear-shaped trophozoite
Leishmania donovani	Kala-azar (sandfly *Phlebotomus*); LD bodies (amastigotes) in splenic/bone-marrow aspirate; rK39 test; NNN medium
Toxoplasma gondii	Cat = definitive host; congenital (chorioretinitis, hydrocephalus, intracranial calcification)
Trichomonas vaginalis	Frothy greenish discharge, 'strawberry cervix'; jerky motility on wet mount
Cryptosporidium	Diarrhoea in AIDS; modified ZN-positive oocysts

Helminths

Worm	Key fact
Ascaris lumbricoides	Commonest worm; Löffler syndrome (lung migration, eosinophilia); intestinal obstruction
Ancylostoma duodenale / Necator	Skin penetration; **iron-deficiency anaemia**
Enterobius vermicularis	Perianal itch at night; **NIH swab / cellophane tape** test
Wuchereria bancrofti	Lymphatic filariasis (Culex); **nocturnal periodicity** — night blood smear 10 pm–2 am; DEC; ICT card test
Taenia solium	Pork tapeworm; **neurocysticercosis** — commonest cause of acquired epilepsy in India
Echinococcus granulosus	Hydatid cyst (dog = definitive host, man = accidental intermediate); liver > lung; Casoni test (old)
Strongyloides	Autoinfection, hyperinfection in steroid users
Schistosoma haematobium	Haematuria; bladder squamous carcinoma

Eosinophilia + worm: think of tissue-migrating helminths (Ascaris, hookworm, filaria, Strongyloides).

Fungi

Fungus	Key fact
Dermatophytes (Trichophyton, Microsporum, Epidermophyton)	Tinea (ringworm); KOH mount; Wood's lamp fluorescence for Microsporum
Candida albicans	Oral thrush (white patches that wipe off), vaginitis; **germ-tube test** positive; pseudohyphae
Cryptococcus neoformans	Meningitis in AIDS; **India ink** capsule; pigeon droppings; latex antigen test
Aspergillus	Fungus ball in old TB cavity; ABPA in asthmatics; **acute-angle (45°) septate branching**
Mucor / Rhizopus	Rhinocerebral mucormycosis in diabetic ketoacidosis (post-COVID surge); **broad aseptate, right-angle branching**
Pneumocystis jirovecii	Pneumonia in AIDS (CD4 < 200); silver stain; co-trimoxazole
Histoplasma	Intracellular yeasts in macrophages
Sporothrix	Rose gardener's disease; lymphocutaneous nodules
Malassezia furfur	Pityriasis versicolor; 'spaghetti and meatballs' on KOH
Rhinosporidium seeberi	Nasal polyps (strawberry-like), common in India; pond bathing

Memory hook: **V**ivax and **O**vale come back ("**VO**lunteer to return"): their liver hypnozoites cause relapses. **Falciparum** is the killer, with banana-shaped gametocytes and many rings in one cell.

Haematology: CBC, anaemia, ESR, coagulation, blood groups

Learn one normal value per test and the anaemia table; most haematology MCQs are built on these.

Normal values (adult)

Test	Normal
Haemoglobin	Men 13–17 g/dL; women 12–15 g/dL; WHO anaemia cut-off in pregnancy < 11 g/dL
RBC count	4.5–5.5 million/µL
Total leucocyte count	4,000–11,000/µL

Test	Normal
Differential	Neutrophils 40–75%, lymphocytes 20–40%, monocytes 2–10%, eosinophils 1–6%, basophils < 1%
Platelets	1.5–4.5 lakh/µL
MCV	80–100 fL
MCH / MCHC	27–32 pg / 32–36 g/dL
Haematocrit (PCV)	Men 40–50%, women 36–46%
Reticulocytes	0.5–2.5%
ESR (Westergren)	Men 0–15, women 0–20 mm in 1st hour
Bleeding time	2–7 min (Ivy)
Clotting time	4–9 min (Lee–White)
PT / INR	11–16 s / ~1.0
aPTT	25–40 s

ESR: raised in TB, rheumatoid arthritis, multiple myeloma (very high, with rouleaux), pregnancy, anaemia, malignancy, temporal arteritis. **Low in polycythaemia, sickle cell disease, CHF, hypofibrinogenaemia**. Wintrobe and Westergren methods; anticoagulant for ESR = 3.8% trisodium citrate (1:4).

Anticoagulant tubes: **EDTA (lavender)** = CBC; **sodium citrate (blue)** = PT/aPTT (1:9); **fluoride (grey)** = glucose (inhibits glycolysis); heparin (green) = some biochemistry and blood gases; plain/red = serum.

Anaemias

Type	Indices and smear	Key tests
Iron deficiency (commonest worldwide and in India)	Microcytic hypochromic; pencil cells; **koilonychia**	Low ferritin (best single test), high TIBC, low serum iron; Mentzer index > 13
Thalassaemia trait	Microcytic, target cells, high RBC count	**HbA2 > 3.5%** on electrophoresis/HPLC; Mentzer < 13; normal ferritin

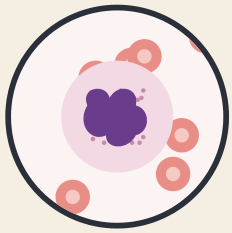
Type	Indices and smear	Key tests
Anaemia of chronic disease	Normo- or microcytic	High ferritin, low TIBC (hepcidin)
Sideroblastic	Microcytic	Ring sideroblasts (Prussian blue on marrow)
Megaloblastic (B12/folate)	**Macrocytic, hypersegmented neutrophils (> 5 lobes)**, macro-ovalocytes	Low B12 / folate; B12 deficiency = subacute combined degeneration; vegans (MO 2025)
Haemolytic	Normocytic; high reticulocytes, unconjugated bilirubin, LDH; low haptoglobin	Coombs test (autoimmune); spherocytes in hereditary spherocytosis (osmotic fragility raised)
Sickle cell (HbS: glutamate → valine at β6)	Sickle cells; Howell–Jolly bodies after autosplenectomy	Sickling test (sodium metabisulphite); HPLC; tribal belts of central and western India
G6PD deficiency	Bite cells, **Heinz bodies** (supravital stain) after oxidant drugs (primaquine, dapsone) or fava beans	G6PD assay
Aplastic	Pancytopenia	Hypocellular bone-marrow **biopsy** (trephine)

Red-cell inclusions: Howell–Jolly (DNA remnants, asplenia); basophilic stippling (lead poisoning, thalassaemia); Pappenheimer (iron); Heinz (denatured Hb); Cabot rings.

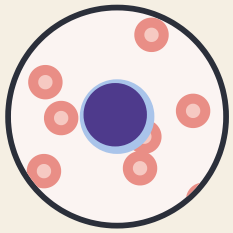
Reading a blood film

Normal white cells, and what anaemia does to red cells

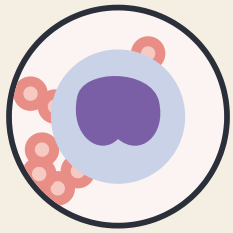
WHITE CELLS · normal differential count



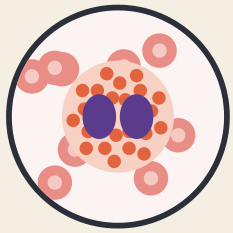
Neutrophil
40–75%
3–5 lobes; leads acute inflammation. Neutrophilia: bacteria



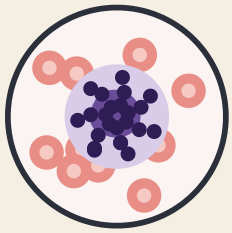
Lymphocyte
20–40%
Round nucleus fills the cell. Raised in viruses, TB, pertussis



Monocyte
2–10%
Largest white cell; kidney-shaped nucleus



Eosinophil
1–6%
Two lobes, red granules. Eosinophilia: allergy, parasites



Basophil
under 1%
Dark granules hide the nucleus; rarest white cell

RED CELLS · what anaemia does to their shape



Normal
Biconcave disc, pale centre



Microcytic, hypochromic
Iron deficiency: small, pale, pencil cells



Target cells
Thalassaemia trait: Mentzer index < 13



Macro-ovalocytes
B12 / folate; with neutrophils > 5 lobes



Spherocytes
Hereditary spherocytosis: no pale centre



Sickle cells
HbS: glutamate → valine at β6



Bite cells
G6PD deficiency after oxidant drugs, fava beans

Inclusions
Howell–Jolly: asplenia
Basophilic stippling: lead, thalassaemia
Heinz bodies: G6PD
Ring sideroblasts: sideroblastic (marrow)

Blood film · normal white cells and the red-cell changes of anaemia

White cells and leukaemia

Disease	Key finding
AML	Myeloblasts with **Auer rods**; MPO positive; M3 (APL) t(15;17), DIC, treat with ATRA
ALL	Commonest childhood cancer/leukaemia; lymphoblasts; PAS positive; CD10 (CALLA)

Disease	Key finding
CML	**Philadelphia chromosome t(9;22), BCR–ABL**; very high TLC with all stages of maturation, basophilia; low LAP score; imatinib
CLL	Elderly; **smudge (basket) cells**; CD5 + CD23 B cells
Hodgkin lymphoma	**Reed–Sternberg cells** (owl's eye, CD15 + CD30); nodular sclerosis commonest
Multiple myeloma	Bone pain, hypercalcaemia, renal failure, anaemia (CRAB); **M-spike** on electrophoresis, **Bence Jones proteins** in urine, punched-out lytic skull lesions, rouleaux

Leukaemoid reaction vs CML: leukaemoid has **high LAP score**, toxic granules, no Philadelphia chromosome.

Bleeding and clotting

Test	Prolonged in
Bleeding time	Platelet disorders: thrombocytopenia (ITP, dengue), von Willebrand disease, aspirin
PT / INR (extrinsic: factor VII)	Warfarin therapy (monitored by INR), liver disease, vitamin K deficiency, DIC
aPTT (intrinsic: VIII, IX, XI, XII)	Haemophilia A (VIII) and B (IX), heparin (monitored by aPTT), von Willebrand, DIC

Haemophilia: X-linked recessive; normal bleeding time and PT, prolonged aPTT; haemarthrosis. **ITP**: low platelets, normal PT/aPTT, megakaryocytes increased in marrow. Vitamin K-dependent factors: **II, VII, IX, X** (and proteins C, S).

Blood groups and transfusion

- ABO antibodies are naturally occurring IgM; Rh antibodies are IgG (cross the placenta).
- **Universal donor = O negative** (red cells); **universal recipient = AB positive**. Universal plasma donor = AB.
- Most dangerous acute transfusion reaction = ABO incompatibility (acute intravascular haemolysis, type II). Commonest = febrile non-haemolytic reaction.
- Storage: whole blood/PRBC 2–6 °C (CPDA-1, 35 days; SAGM, 42 days); **platelets 20–24 °C with agitation, 5 days**; FFP ≤ −30 °C (1 year). Cryoprecipitate = fibrinogen, factor VIII, vWF, XIII.

- Rh-negative mother: **anti-D** at 28 weeks and within 72 hours of delivery of an Rh-positive baby; indirect Coombs test to monitor.

Memory hook: tube colours: **L**avender for the b**L**ood count (EDTA), **B**lue for **B**leeding tests (citrate), **G**rey for **G**lucose (fluoride). Small red cells are **TICS**: **T**halassaemia, **I**ron deficiency, **C**hronic disease, **S**ideroblastic.

Clinical pathology: LFT, RFT, urine, CSF, glucose, lipids, enzymes

Questions here are pattern-recognition: which enzyme rises, which jaundice, which CSF picture.

Liver function tests

Test	Normal	Raised in
Total bilirubin	0.3–1.2 mg/dL (clinical jaundice > 2–3)	Jaundice
ALT (SGPT)	< 40 U/L	**Hepatocellular damage (most liver-specific)**; viral hepatitis ALT > AST
AST (SGOT)	< 40 U/L	Liver, heart, muscle; **AST:ALT > 2 = alcoholic liver disease**
Alkaline phosphatase	40–130 U/L	**Cholestasis / obstruction**, bone disease (Paget's, rickets, metastases), pregnancy, growing children
GGT	—	Alcohol, cholestasis (confirms hepatic origin of ALP)
Albumin	3.5–5 g/dL	Low in chronic liver disease, nephrotic syndrome, malnutrition (synthetic function)
PT/INR	—	Prolonged = liver synthetic failure (best acute marker)

Jaundice patterns

Type	Bilirubin	Urine	Stool
Pre-hepatic (haemolytic)	Unconjugated ↑	Urobilinogen ↑, no bilirubin (acholuric)	Dark
Hepatic	Both ↑	Bilirubin +, urobilinogen variable	Normal/pale

Type	Bilirubin	Urine	Stool
Post-hepatic (obstructive)	Conjugated ↑, ALP ↑↑	**Bilirubin +, urobilinogen absent**; dark urine	**Clay-coloured**; itching

Three jaundices, told apart by urine, stool and enzymes

Read across a row to compare; read down a column to recall one type

	Pre-hepatic (haemolytic)	Hepatic (liver cell)	Post-hepatic (obstructive)
Bilirubin raised	unconjugated	both kinds	conjugated
Urine bilirubin	absent (acholuric)	present	present: dark urine
Urobilinogen	increased	normal or raised	absent
Stool	normal or dark	normal to pale	pale, clay-coloured
Key test	reticulocytes, LDH up	ALT, AST very high	ALP, GGT very high

jaundice types · pre-hepatic, hepatic, obstructive

The right-hand column is the one asked most: dark urine with pale stools means the bile cannot reach the gut.

Gilbert syndrome = mild unconjugated rise with fasting/stress; Crigler–Najjar = UGT deficiency; Dubin–Johnson = conjugated, black liver.

Renal function tests

- Blood urea 15–40 mg/dL (BUN 7–20); serum creatinine 0.6–1.2 mg/dL. Creatinine is better than urea; **eGFR** (from creatinine) staging CKD.
- Urea:creatinine ratio > 20 = pre-renal (dehydration, GI bleed).
- GFR ~125 mL/min; **inulin clearance = gold standard GFR**; creatinine clearance in practice; PAH = renal plasma flow.
- Uric acid: men 3.5–7.2 mg/dL; raised in gout, tumour lysis, leukaemia, thiazides.
- Electrolytes: Na 135–145, K 3.5–5.0 mmol/L. **Hyperkalaemia ECG: tall peaked T waves**; hypokalaemia: U waves.

Urine examination

Finding	Meaning
Normal output 1–2 L/day; oliguria < 400 mL; anuria < 100 mL; polyuria > 3 L	—
Specific gravity 1.003–1.030; fixed 1.010 (isosthenuria)	Chronic renal failure
Proteinuria > 3.5 g/day	**Nephrotic syndrome** (with oedema, hypoalbuminaemia, hyperlipidaemia, lipiduria — oval fat bodies, Maltese cross)
Microalbuminuria 30–300 mg/day	Earliest sign of diabetic nephropathy
RBC casts, dysmorphic RBCs	Glomerulonephritis (nephritic syndrome)
WBC casts, pus cells	Pyelonephritis
Granular / muddy-brown casts	Acute tubular necrosis
Hyaline casts	Normal / concentrated urine
Bence Jones protein (heat test: precipitates at 40–60 °C, redissolves at 100 °C)	Multiple myeloma
Ketones (Rothera's test)	DKA, starvation
Glucose (Benedict's test)	Glycosuria above renal threshold (~180 mg/dL)
Significant bacteriuria ≥ 10^5 CFU/mL (midstream urine)	UTI (E. coli commonest)
Haemoglobinuria vs haematuria	Dipstick positive without RBCs = haemoglobin/myoglobin

Crystals: calcium oxalate (envelope), uric acid (rhomboid), triple phosphate (coffin-lid, Proteus infection), cystine (hexagonal).

CSF

Condition	Cells	Protein	Glucose	Clue
Normal	0–5 lymphocytes	15–45 mg/dL	40–70 mg/dL (≈ 2/3 blood)	Clear, opening pressure 5–18 cmH2O
Bacterial (pyogenic)	Neutrophils ↑↑	↑↑	↓↓	Turbid; Gram stain

Condition	Cells	Protein	Glucose	Clue
Tuberculous	Lymphocytes ↑	↑↑	↓	**Cobweb clot** on standing; ADA ↑
Viral	Lymphocytes ↑	Normal/slightly ↑	**Normal**	Clear
Fungal (Cryptococcus)	Lymphocytes	↑	↓	India ink
Subarachnoid haemorrhage	RBCs	↑	Normal	**Xanthochromia**
Guillain–Barré	Normal	↑↑	Normal	**Albuminocytological dissociation**

Glucose and diabetes

- Diabetes: fasting plasma glucose ≥ 126 mg/dL, 2-h OGTT ≥ 200, random ≥ 200 with symptoms, or **HbA1c ≥ 6.5%**. Prediabetes: FPG 100–125 (IFG); 2-h 140–199 (IGT); HbA1c 5.7–6.4%.
- HbA1c reflects the average glucose of the previous ~**2–3 months** (RBC life 120 days). Fructosamine = 2–3 weeks.
- Type 1 = ketoacidosis-prone (MO 2025 trap: type 2 is *not* more ketosis-prone); type 2 deaths are mostly cardiovascular.
- Gestational diabetes in India: single-step 75 g OGTT (DIPSI), 2-h ≥ 140 mg/dL.

Lipids: total cholesterol < 200, **LDL < 100** (optimal), HDL > 40 (men) / > 50 (women), triglycerides < 150 mg/dL. Friedewald: LDL = TC − HDL − TG/5.

Enzymes and markers

Marker	Use
Troponin I/T	Myocardial infarction — most sensitive and specific; rises 3–4 h, stays 7–10 days
CK-MB	MI; returns to normal by 48–72 h (good for re-infarction)
LDH (LDH1 > LDH2 'flipped')	Late MI marker; haemolysis, lymphoma
Amylase, **lipase** (more specific)	Acute pancreatitis (> 3× normal)

Marker	Use
Acid phosphatase / PSA	Prostate
ALP	Bone, cholestasis
ADA	TB effusions
CRP, procalcitonin	Inflammation; procalcitonin = bacterial sepsis
BNP / NT-proBNP	Heart failure
D-dimer	DVT / PE exclusion; DIC
TSH	Best screening test for thyroid function; low TSH + high T4 = hyperthyroidism (Graves', MO 2024)

Body fluids: pleural fluid — exudate by Light's criteria (see haemodynamics); ADA > 40 U/L suggests TB pleurisy; semen analysis (WHO 2021 lower limits): volume 1.4 mL, count 16 million/mL, progressive motility 30%, normal forms 4%.

Memory hook: obstructive jaundice is **dark urine, pale stools, itching and a very high ALP**. In CSF, **bacteria eat the sugar** (low glucose, neutrophils); **viruses leave it alone** (normal glucose, lymphocytes); **TB** eats it slowly (low glucose, lymphocytes, cobweb clot).

Imaging and serology: X-ray, ultrasound, markers

GPSC asks named radiological signs and 'investigation of choice' — both appeared in the 2024 and 2025 Medical Officer papers.

Chest X-ray

Sign / finding	Meaning
Right heart border = SVC, right atrium (IVC low)	Right ventricle is NOT on the right border (MO 2024); left border = aortic knuckle, pulmonary trunk, left auricle, left ventricle
Cardiothoracic ratio > 0.5 (PA view)	Cardiomegaly
Air under the diaphragm (erect film)	Perforated hollow viscus (peptic ulcer)

Sign / finding	Meaning
Blunting of costophrenic angle; meniscus sign	Pleural effusion (≥ 200–300 mL on PA view; lateral decubitus detects less)
Air bronchogram, lobar opacity	Consolidation (pneumonia)
Apical cavity, fibrosis	Post-primary TB
Millet-seed shadows	Miliary TB
Kerley B lines, bat-wing opacity, upper-lobe diversion	Pulmonary oedema / left heart failure
Hyperinflation, flat diaphragm	COPD/emphysema
Visceral pleural line, no lung markings	Pneumothorax
Westermark sign, Hampton's hump	Pulmonary embolism (CT pulmonary angiography = investigation of choice)
Egg-shell calcification of hilar nodes	Silicosis (also sarcoidosis)
Boot-shaped heart	Tetralogy of Fallot
Egg-on-string	Transposition of great arteries
Snowman / figure-of-8	TAPVC

Other X-ray and barium signs

Sign	Condition
Bird's beak	Achalasia cardia
Rat tail / shouldering	Carcinoma oesophagus (MO 2025)
Apple-core lesion	Colorectal carcinoma
String sign of Kantor	Crohn's disease
Lead-pipe colon	Ulcerative colitis
Coffee-bean sign	Sigmoid volvulus

Sign	Condition
Double-bubble	Duodenal atresia
Step-ladder fluid levels	Small-bowel obstruction
Cupping, fraying, splaying of metaphysis	Rickets (sub-periosteal haemorrhage = scurvy, MO 2024)
Hair-on-end skull	Thalassaemia, sickle cell
Pepper-pot skull	Hyperparathyroidism
Punched-out lytic skull lesions	Multiple myeloma
Sun-ray appearance, Codman's triangle	Osteosarcoma
Onion-peel periosteal reaction	Ewing sarcoma
Soap-bubble appearance	Giant-cell tumour (osteoclastoma)
Bamboo spine	Ankylosing spondylitis

Ultrasound and other imaging

Finding	Meaning
Ultrasound = investigation of choice	Gallstones (acoustic shadow), obstetrics, renal calculi screening, liver abscess
Snowstorm appearance	Hydatidiform mole
Bagel / tubal ring sign	Ectopic pregnancy (MO 2024)
String of pearls ovaries	PCOS
Target / pseudokidney sign	Intussusception
Water-lily sign	Hydatid cyst (collapsed membrane)
Investigation of choice: acute subarachnoid haemorrhage	**Non-contrast CT** (MO 2025)
Intraocular metallic foreign body	CT orbit; MRI contraindicated (MO 2025)
Spine TB / cord compression imaging	MRI (diagnosis confirmed by biopsy)
Pulmonary embolism	CT pulmonary angiography

Finding	Meaning
Bronchiectasis	HRCT
Renal stones	Non-contrast CT (KUB)
Ventricular function	Echocardiography

Serological markers (rapid recall)

Disease	Marker / test
Typhoid	Blood culture (week 1), Widal (week 2), Typhidot (IgM)
Dengue	NS1 (day 1–5), IgM ELISA (after day 5)
Malaria	Smear (gold standard); RDT: HRP-2 (falciparum), pLDH
Leptospirosis	IgM ELISA; MAT (reference)
Scrub typhus	IgM ELISA; Weil–Felix OX-K
HIV	ELISA/rapid tests × 3 (India); DNA PCR in infants
Hepatitis B	HBsAg (screen); anti-HBc IgM (acute); anti-HBs (immunity)
Syphilis	VDRL/RPR (screen, follow-up); TPHA (confirm)
Rheumatic fever	ASO titre; Jones criteria
Rheumatoid arthritis	Anti-CCP, RF
SLE	ANA (screen), anti-dsDNA (specific)
Infectious mononucleosis	Paul–Bunnell / Monospot
Kala-azar	rK39 rapid test
Filariasis	Night smear; ICT card for antigen
COVID-19	RT-PCR on nasopharyngeal swab

Memory hook: X-ray pictures have names: **millet seeds** = miliary TB, **bat wings** = pulmonary oedema, **Kerley B lines** = interstitial oedema, **steeple sign** = croup, **snowstorm** on ultrasound = molar pregnancy.

Rapid revision: 60 one-liners

Read this page the night before and again on the morning of the exam.

'Most common' and 'gold standard'

1. Commonest type of necrosis: coagulative. Brain: liquefactive.
2. Commonest embolism: pulmonary thromboembolism from deep leg veins.
3. Commonest inherited thrombophilia: factor V Leiden.
4. Commonest mutated gene in cancer: p53; commonest oncogene: RAS.
5. Commonest site of metastasis: liver (then lungs); lymph nodes for carcinomas.
6. Commonest anaemia: iron deficiency; best test serum ferritin.
7. Commonest childhood malignancy: ALL.
8. Commonest cause of infantile diarrhoea: rotavirus.
9. Commonest cause of acquired epilepsy in India: neurocysticercosis.
10. Commonest UTI organism: E. coli.
11. Commonest site of amoebic liver abscess: right lobe.
12. Commonest worm: Ascaris.
13. Gold standard for malaria: peripheral smear. For TB: culture. For GFR: inulin clearance. For amyloid: Congo red with apple-green birefringence.
14. Investigation of choice: gallstones = ultrasound; SAH = non-contrast CT; PE = CT pulmonary angiography; bronchiectasis = HRCT.

Pairs (stain, medium, test)

15. ZN stain → AFB; Albert → diphtheria granules; India ink → Cryptococcus; Giemsa → malaria.
16. Congo red → amyloid; Perls (Prussian blue) → iron; von Kossa → calcium; PAS → glycogen, fungi; Sudan/Oil Red O → fat.
17. LJ → TB; TCBS → cholera; Löffler/tellurite → diphtheria; Sabouraud → fungi; EMJH → Leptospira; NNN → Leishmania.
18. Autoclave 121 °C, 15 psi, 15 min; hot-air oven 160 °C, 2 h.
19. Spore indicators: autoclave = Geobacillus stearothermophilus; hot-air oven = Bacillus atrophaeus.
20. Endoscopes: 2% glutaraldehyde. Plastics: ethylene oxide. Disposable syringes: gamma rays. Sera: filtration.
21. EDTA → CBC; citrate → PT/aPTT and ESR; fluoride → glucose.

Bodies, cells and signs

22. Lines of Zahn → ante-mortem thrombus.
23. Langhans giant cells → TB; Reed–Sternberg → Hodgkin; Aschoff bodies → rheumatic fever.
24. Negri bodies → rabies; owl's-eye inclusions → CMV; Councilman bodies → viral hepatitis (apoptosis); Mallory bodies → alcoholic hepatitis.
25. Auer rods → AML; smudge cells → CLL; Philadelphia chromosome → CML.
26. Psammoma bodies → papillary thyroid, serous ovarian, meningioma.
27. Koilocytes → HPV; Tzanck giant cells → herpes/varicella.
28. Crescent gametocytes → falciparum; Schüffner's dots → vivax.
29. Hypersegmented neutrophils → megaloblastic anaemia; target cells → thalassaemia, liver disease; Heinz bodies → G6PD deficiency; Howell–Jolly → asplenia.
30. Troisier's / Virchow's node → gastric carcinoma; Krukenberg → stomach to ovary.
31. Conjunctival suffusion + calf pain → leptospirosis; eschar → scrub typhus; rice-water stool → cholera.
32. Anchovy-sauce pus → amoebic liver abscess; cobweb CSF → TB meningitis; xanthochromia → SAH.

Numbers

33. Fasting glucose ≥ 126 mg/dL; HbA1c ≥ 6.5% = diabetes; HbA1c shows 2–3 months.
34. Haemoglobin: anaemia < 13 (men), < 12 (women), < 11 (pregnancy).
35. TLC 4,000–11,000/µL; platelets 1.5–4.5 lakh/µL.
36. Serum creatinine 0.6–1.2 mg/dL; urea 15–40 mg/dL.
37. Nephrotic proteinuria > 3.5 g/day; microalbuminuria 30–300 mg/day.
38. Significant bacteriuria ≥ 10^5 CFU/mL.
39. HbA2 > 3.5% → β-thalassaemia trait.
40. Mantoux ≥ 10 mm positive, read at 48–72 h.
41. Transudate protein < 3 g/dL; exudate > 3 g/dL.
42. Platelets stored at 20–24 °C (5 days); PRBC 2–6 °C.
43. AIDS: CD4 < 200/µL.
44. Rising haematocrit ≥ 20% in dengue = plasma leakage.

Classic traps

45. Lymphoma, melanoma, seminoma, mesothelioma are malignant despite '-oma'.
46. Typhoid ulcers are longitudinal; TB ulcers are transverse.

47. ESR is LOW in polycythaemia and sickle cell disease.
48. Hepatitis B is a DNA virus; A, C, D, E are RNA.
49. HEV kills pregnant women; HCV goes chronic most often.
50. IgM is the first antibody; IgG alone crosses the placenta; IgA is in secretions.
51. Type 1 diabetes, not type 2, is ketosis-prone.
52. Stage, not grade, predicts prognosis.
53. Mantoux and lepromin test cell-mediated (type IV) immunity; lepromin gives prognosis, not diagnosis.
54. Bleeding time is normal in haemophilia; aPTT is prolonged.
55. PT/INR monitors warfarin; aPTT monitors heparin.
56. Red infarcts in lung and liver (dual supply); white infarcts in heart, kidney, spleen.
57. Prostate bone metastases are osteoblastic.
58. Paget disease of the nipple is malignant.
59. Little's area is anteroinferior on the nasal septum (GPSC's 2024 key is wrong).
60. Mosquito is the definitive host of malaria; man is the intermediate host.

One-glance recall card

Topic 6 recall card: the 15 facts asked most

Cover the small lines, say the fact from the big cue, then check

P-R-L nucleus dies: pyknosis, karyorrhexis, karyolysis	**Roll vs stop** selectins roll the cell; integrins stop it	**ISH** Virchow: injury, stasis, hypercoagulability
Grade vs stage stage (spread) predicts outcome better than grade	**ACID** I IgE, II cytotoxic, III complex, IV T cell	**IgG** the only antibody that crosses the placenta
121 · 15 · 15 autoclave: °C, lb/sq in, minutes; kills spores	**BASU** typhoid: blood, antibody, stool, urine by week	**Vowels to bowels** hepatitis A and E spread faecal-oral; B, C, D by blood
Negri bodies rabies; Guarnieri = vaccinia, owl's eye = CMV	**Vivax, Ovale** hypnozoites cause relapse; falciparum kills	**TICS** microcytic: thalassaemia, iron, chronic, sideroblastic
Lavender tube EDTA for blood count; blue citrate; grey fluoride	**Dark urine, pale stool** obstructive jaundice, ALP very high	**CSF sugar low** bacterial or TB; normal in viral

Topic 6 recall card · 15 tiles

For the last ten minutes before the exam: cover the small lines of each tile and say the fact from the big cue.

Practice MCQs (50)

Time yourself: 50 questions in 35 minutes. Cover the key below until you finish.

1. The earliest change of irreversible cell injury seen on light microscopy is: (A) nuclear pyknosis (B) cell swelling (C) fatty change (D) loss of microvilli
2. Caseous necrosis is typical of: (A) myocardial infarction (B) acute pancreatitis (C) brain infarct (D) tuberculosis
3. Fat necrosis is characteristic of: (A) TB (B) gangrene (C) acute pancreatitis (D) malignant hypertension
4. Apoptosis differs from necrosis because it: (A) causes inflammation (B) affects sheets of cells (C) always follows ischaemia (D) shows cell shrinkage without inflammation

5. Squamous change in the bronchus of a smoker is: (A) dysplasia (B) hyperplasia (C) metaplasia (D) anaplasia
6. Dystrophic calcification occurs with: (A) normal serum calcium in damaged tissue (B) high serum calcium (C) vitamin D excess (D) hyperparathyroidism
7. Psammoma bodies are seen in: (A) follicular carcinoma thyroid (B) medullary carcinoma thyroid (C) anaplastic carcinoma thyroid (D) papillary carcinoma thyroid
8. The chief chemotactic factor among these is: (A) C5a (B) C3b (C) histamine (D) bradykinin
9. The cell typical of a granuloma is the: (A) epithelioid cell (B) neutrophil (C) eosinophil (D) mast cell
10. Langhans giant cells have nuclei arranged: (A) scattered (B) at the periphery in a horseshoe (C) in the centre (D) around a lipid core
11. The hallmark of healing is: (A) fibrin (B) pus (C) granulation tissue (D) necrosis
12. Lines of Zahn indicate: (A) a post-mortem clot (B) an infarct (C) an ante-mortem thrombus (D) a fat embolus
13. The commonest source of pulmonary emboli is: (A) right atrium (B) pelvic veins only (C) deep veins of the legs (D) arm veins
14. A red (haemorrhagic) infarct is typical of the: (A) lung (B) heart (C) kidney (D) spleen
15. Fat embolism most often follows: (A) fracture of long bones (B) caisson disease (C) labour (D) IV injection
16. Neutrophils dominate a myocardial infarct at: (A) 0–4 hours (B) 7–10 days (C) 1–3 days (D) 2 months
17. Which of these is malignant? (A) hamartoma (B) choristoma (C) lipoma (D) seminoma
18. The best predictor of prognosis in cancer is: (A) stage (B) grade (C) mitotic count (D) tumour marker level
19. Aflatoxin is linked to carcinoma of the: (A) stomach (B) liver (C) bladder (D) lung
20. A Krukenberg tumour is a metastasis to the ovary usually from the: (A) breast (B) colon (C) uterus (D) stomach
21. CA-125 is a marker for carcinoma of the: (A) pancreas (B) colon (C) prostate (D) ovary
22. Amyloid shows apple-green birefringence with: (A) PAS (B) Perls stain (C) Congo red under polarised light (D) von Kossa
23. The tuberculin (Mantoux) reaction is hypersensitivity type: (A) I (B) IV (C) II (D) III
24. The immunoglobulin that crosses the placenta is: (A) IgA (B) IgM (C) IgE (D) IgG
25. Standard autoclave conditions are: (A) 100 °C for 30 min (B) 160 °C for 1 h (C) 60 °C for 30 min (D) 121 °C, 15 psi, 15 min
26. Endoscopes are best disinfected with: (A) autoclaving (B) hot-air oven (C) 2% glutaraldehyde (D) boiling
27. India ink preparation is used to detect: (A) Candida (B) Cryptococcus (C) Aspergillus (D) Mucor
28. Metachromatic granules are characteristic of: (A) Mycobacterium (B) Corynebacterium diphtheriae (C) Clostridium (D) Streptococcus
29. TCBS medium is used to culture: (A) Vibrio cholerae (B) Salmonella (C) Shigella (D) E. coli

30. In the first week of typhoid the most useful test is: (A) Widal (B) blood culture (C) urine culture (D) stool culture
31. Typhoid ulcers in the ileum are: (A) longitudinal (B) transverse (C) circular girdle (D) flask-shaped
32. Leptospirosis is mainly acquired through: (A) mosquito bite (B) droplets (C) blood transfusion (D) water or soil contaminated with rat urine
33. An eschar with fever after field work suggests: (A) scrub typhus (B) leptospirosis (C) dengue (D) malaria
34. The window period of acute hepatitis B is detected by: (A) HBsAg (B) anti-HBs (C) anti-HBc IgM (D) HBeAg
35. The hepatitis with the highest mortality in pregnancy is: (A) hepatitis A (B) hepatitis B (C) hepatitis C (D) hepatitis E
36. Dengue on day 2 of fever is best confirmed by: (A) IgG ELISA (B) NS1 antigen (C) Widal (D) IgM ELISA
37. Negri bodies are diagnostic of: (A) measles (B) CMV (C) rabies (D) herpes
38. Banana-shaped gametocytes are seen in: (A) P. vivax (B) P. falciparum (C) P. malariae (D) P. ovale
39. Relapse in vivax malaria is due to: (A) hypnozoites (B) gametocytes (C) merozoites (D) sporozoites in blood
40. Anchovy-sauce pus is typical of: (A) pyogenic liver abscess (B) amoebic liver abscess (C) hydatid cyst (D) hepatoma
41. A night blood smear is needed for: (A) malaria (B) filariasis (C) kala-azar (D) typhoid
42. The commonest anaemia in India is: (A) megaloblastic (B) sickle cell (C) aplastic (D) iron deficiency
43. Hypersegmented neutrophils suggest: (A) iron deficiency (B) thalassaemia (C) vitamin B12 or folate deficiency (D) lead poisoning
44. HbA2 above 3.5% indicates: (A) iron deficiency (B) sickle cell anaemia (C) β-thalassaemia trait (D) aplastic anaemia
45. ESR is LOW in: (A) polycythaemia (B) TB (C) multiple myeloma (D) pregnancy
46. In haemophilia A, the test that is prolonged is: (A) bleeding time (B) PT (C) thrombin time (D) aPTT
47. The Philadelphia chromosome t(9;22) is the hallmark of: (A) AML (B) CML (C) CLL (D) ALL
48. Absent urinary bilirubin with raised urobilinogen suggests: (A) obstructive jaundice (B) haemolytic jaundice (C) hepatitis (D) Dubin–Johnson syndrome
49. CSF with lymphocytes, high protein, low sugar and a cobweb clot suggests: (A) pyogenic meningitis (B) tuberculous meningitis (C) viral meningitis (D) subarachnoid haemorrhage
50. The most specific marker of myocardial infarction is: (A) troponin (B) CK total (C) LDH (D) AST

Answer key

Q	Ans	Why
1	A	Nuclear changes mark irreversibility

Q	Ans	Why
2	D	Cheesy necrosis in granulomas
3	C	Lipase causes saponification
4	D	No inflammation, single cells
5	C	Columnar to squamous
6	A	Dead tissue, normal calcium
7	D	Papillary thyroid carcinoma
8	A	C5a (also LTB4, IL-8)
9	A	Epithelioid cell = activated macrophage
10	B	Horseshoe at the periphery
11	C	New capillaries + fibroblasts
12	C	Formed in flowing blood
13	C	Over 95% from deep leg veins
14	A	Dual blood supply (lung, liver)
15	A	Long-bone fractures
16	C	Days 1 to 3
17	D	Seminoma is malignant despite '-oma'
18	A	TNM stage
19	B	Hepatocellular carcinoma
20	D	Signet-ring gastric carcinoma
21	D	Epithelial ovarian cancer
22	C	Congo red, polarised light
23	B	Delayed, T-cell mediated
24	D	IgG only
25	D	121 °C, 15 psi, 15–20 min

Q	Ans	Why
26	C	Cidex
27	B	Capsule in CSF
28	B	Volutin granules (Albert stain)
29	A	Yellow colonies
30	B	Bacteraemia in week 1
31	A	Along Peyer's patches
32	D	Rat urine in water or soil
33	A	Chigger (larval mite) bite
34	C	Only marker in the window
35	D	About 20% mortality in pregnancy
36	B	NS1 positive in days 1–5
37	C	Hippocampus and Purkinje cells
38	B	Crescent gametocytes
39	A	Dormant liver stage
40	B	Entamoeba histolytica
41	B	Nocturnal periodicity
42	D	Low ferritin
43	C	Megaloblastic anaemia
44	C	HPLC / electrophoresis
45	A	Also sickle cell disease, CHF
46	D	Factor VIII is intrinsic pathway
47	B	BCR–ABL fusion
48	B	Unconjugated bilirubin is not filtered
49	B	Tuberculous meningitis

Q	Ans	Why
50	A	Troponin I or T

Score guide: 40+ is exam-ready; below 30, reread the tables for the questions you missed.

Sources

GPSC question papers with answer keys (read in full; questions quoted in shortened form):

- Medical Officer CBRT, 28 Sep 2025
- Medical Officer CBRT, 28 Sep 2024
- Public Health Dentist CBRT, 18 Feb 2024
- Junior Ophthalmologist (ESI) CBRT, 5 Nov 2023
- GPSC question paper archive

Standard textbooks the notes follow (facts written from them, not copied; check a doubtful point against your own edition):

- Robbins and Cotran, *Pathologic Basis of Disease* — cell injury, inflammation, haemodynamics, neoplasia, amyloid.
- Harsh Mohan, *Textbook of Pathology* — Indian emphasis, haematology, clinical pathology.
- Ananthanarayan and Paniker, *Textbook of Microbiology* — sterilisation, stains, media, bacteria, viruses.
- C. K. Jayaram Paniker, *Textbook of Medical Parasitology* — malaria, protozoa, helminths.
- Godkar, *Textbook of Medical Laboratory Technology* — normal values, urine, CSF, body fluids.

Indian programme rules (NTEP TB testing, dengue and malaria testing, HIV testing strategy, UIP vaccines) are as of mid-2026; confirm any figure that the paper words differently.